1 Appendix: Sensitivity Analysis of Noise Standard Deviation

To evaluate the sensitivity of MERGEvolve to the noise standard deviation, we vary the parameter σ from 0.0001 to 0.1 while keeping all other hyperparameters unchanged. The results are reported in Table A1.

Table A1. Performance of different datasets under varying σ values. Bold numbers indicate the best performance for each dataset.

Dataset	$\sigma = 0.0001$	$\sigma = 0.001$	$\sigma = 0.01$	$\sigma = 0.1$
ARC-C	68.00	**70.47**	69.45	70.39
DROP	**38.30**	36.60	37.70	35.70
MATH	14.70	15.45	14.64	**15.56**
MELD	51.81	**52.85**	51.37	52.21
MMLU	52.10	**54.40**	48.10	54.20

Overall, MERGEvolve exhibits stable performance across a wide range of σ values. Although the optimal value varies slightly across datasets, the performance differences remain relatively small, indicating that the proposed framework is not highly sensitive to the choice of σ.